

Practical Session – 3D Reconstruction with Gaussian Splatting

1 Objectives

The objective of this practical session is to discover a recent method for 3D scene reconstruction and representation: **3D Gaussian Splatting**.

Starting from a set of images of the same scene, you will:

- estimate the camera positions;
- reconstruct an initial 3D representation of the scene;
- train a Gaussian Splatting model;
- generate novel views of the scene;
- analyse the strengths and limitations of the resulting reconstruction.

2 General principle

Consider a set of images captured from different viewpoints

$$\mathcal{I} = \{I_1, I_2, \dots, I_N\}$$

The first step consists in estimating the camera parameters associated with each image:

$$P_i = K_i[R_i | t_i],$$

where K_i contains the intrinsic camera parameters and (R_i, t_i) describes the camera pose in the scene. These parameters can then be used to initialise a 3D representation of the scene. In Gaussian Splatting, the scene is represented by a collection of 3D Gaussian primitives.

Each Gaussian contains, among other parameters:

- a position $\mu \in \mathbb{R}^3$;
- a covariance matrix Σ describing its shape;
- colour information;
- an opacity.

A Gaussian can be described as

$$G(x) = \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right).$$

The parameters of these primitives are optimised so that images rendered from the training cameras reproduce the input photographs.

3 Part 1 – Image acquisition

Choose a relatively static scene or object. Capture between **30 and 100 photographs** from different viewpoints. You can also use one of the following resources:

- [Mip-NeRF 360 Dataset](#) – indoor and outdoor scenes captured from many viewpoints;
- [Tanks and Temples](#) – multi-view image sequences of real scenes and objects;
- [ETH3D](#) – multi-view stereo datasets with calibrated image sequences;
- [3D Gaussian Splatting project page](#) – datasets and resources associated with the original 3D Gaussian Splatting work.

Question

Why is significant overlap between consecutive images important for 3D reconstruction?

4 Part 2 – Camera estimation with COLMAP

We will use **COLMAP** to estimate camera poses and produce an initial sparse 3D reconstruction. COLMAP is a Structure-from-Motion (SfM) and Multi-View Stereo (MVS) reconstruction system. Documentation and tutorials are available at:

<https://colmap.github.io/>

Place your images in a directory organised as follows:

```
dataset/
images/
image001.jpg
image002.jpg
image003.jpg
...
```

Using COLMAP, perform the following steps:

1. feature extraction;
2. feature matching between images;
3. camera pose estimation;
4. sparse 3D reconstruction.

You can use COLMAP’s graphical interface and the **Automatic Reconstruction** functionality. Inspect the resulting reconstruction using the COLMAP viewer.

Questions

1. What do the reconstructed 3D points represent?
2. Why is this reconstruction called *sparse*?
3. Were all input photographs successfully registered?
4. What situations could prevent COLMAP from correctly estimating a camera pose?

5 Part 3 – Gaussian Splatting training

For this part, you will use [gsplat](#), an open-source library for Gaussian Splatting.

Your objective is to install the library, prepare your COLMAP reconstruction in a compatible format, train a Gaussian Splatting representation of your scene, and render the resulting model.

The documentation and examples provided by the project should contain the information necessary to complete this part:

<https://docs.gsplat.studio/>

You are expected to investigate the available examples and determine the appropriate training procedure yourself.

Questions

1. How does the visual quality evolve during training?
2. How does the number of Gaussians evolve?
3. Why do you think some regions require more Gaussians than others?

6 Part 4 – Novel view synthesis

Once training is complete, render images from viewpoints that do not exactly correspond to the original photographs.

Compare:

- a real input image;
- a reconstruction rendered from a nearby camera;
- a novel view located between two training cameras.

Questions

Identify at least three types of visible artefacts in the generated images.

For each artefact, suggest a possible explanation.

Where do these artefacts appear most frequently?

7 Part 5 – Experiment

Choose **one** of the following experiments.

Experiment A – Number of input images

Repeat the reconstruction using only a subset of the original images.

For example:

$$N = 10, \quad N = 30, \quad N = 60.$$

Compare the resulting reconstructions.

What happens when the number of available viewpoints decreases?

Experiment B – Viewpoint coverage

Remove all images corresponding to one side or region of the scene.

Train the reconstruction again and analyse the result.

What happens when rendering the scene from viewpoints that were poorly represented in the training images?

Experiment C – Number of optimisation iterations

Compare the reconstruction at different stages of the optimisation:

$$1000, \quad 7000, \quad 30000$$

iterations.

Describe how the representation evolves.

Experiment D – Challenging scene

Intentionally select a scene containing one or several difficult elements:

- shiny surfaces;
- transparent objects;
- large uniform regions;

- moving elements;
- significant illumination changes.

Describe and explain the reconstruction failures you observe.

8 Analysis

Answer the following questions briefly.

1. Does Gaussian Splatting explicitly reconstruct a geometric surface?
2. What is the difference between:
 - a point cloud;
 - a triangle mesh;
 - a set of 3D Gaussians?
3. Why can Gaussian Splatting produce realistic images from novel viewpoints?
4. What are the main advantages of Gaussian Splatting?
5. What are its main limitations?
6. Is a visually accurate reconstruction necessarily a geometrically accurate reconstruction? Explain.

Going further

Gaussian Splatting can be seen as a representation lying between explicit and implicit 3D representations. This raises an interesting question:

What information about the actual geometry of the scene is encoded by the Gaussians?